

Formatted Representation of Economic Models: An Econ-ARK Initiative

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Computational economic models are often hard-coded. Today, computing methods advance faster than economics research. There is a need to represent models separate from solution techniques. We do that, inspired by Dolang and DYNARE, Our current focus is on heterogenous agent macroeconomics and macrofinance models

1 INTRODUCTION

Econ-ARK is a an open source project that aims to improve reproducibility in economics research and applications.

Currently, most economic models are bespoke and hard-coded. There is not a standard way to represent or compute over models. This makes replication and reproducibility more difficult. Economic models are often more computationally complex than other kinds of simulations because they involve rational actors which typically must solve rather complex optimization problems. This makes tying modeling to computational 'solvers' tempting.

We are designing and implemented a new way of representing economic models. This format contains the data needed to define the model, but is agnostic to the software solver. Importantly, this will allow the same model to be solved using different, comparable, approximate methods. This will enable better bench-marking of computational results within economics research and applications.

We are currently implementing a reference implementation of this language as we approach the 1.0 version of our flagship economics library, HARK. You can find out more about this at <https://econ-ark.org/>.

We are inspired by other open source projects in the Economics space, such as DYNARE and Dolo. However, we aspire to enable the representation of a wider range of models than either of these tools allow.

2 BLOCKS

One principle of the modeling format is *modularity*.

We want users to be able to define elements of models, and recombine them into new ones.

2.1 Example: Consumption

For example, the following is an example of the markup for part of a consumption-savings problem:

```
block1: &consumption
  name: consumption
  shocks:
    theta: !MeanOneLogNormal
      sigma: 0.1
  dynamics:
    b: k * R
    m: b + theta
    c: !Control
      info: m
      upper_bound: m
    a: m - c
  reward:
    u: c ** (1 - CRRA) / (1 - CRRA)
```

This block corresponds to the following equations:

$$\begin{aligned}
 \theta &\sim \text{Lognormal}(1, 0.1) \\
 b &= Rk \\
 m &= b + \theta \\
 a &= m - c \\
 c &< m
 \end{aligned} \tag{1}$$

$$u(c) = \frac{c^{1-\rho}}{1-\rho}$$

It identifies that c is a decision (or control) variable, and the information set for that control is m . Given a decision rule $c(m)$, this system of equations governs the dynamics from k to a .

Notice:

- This is a flavor of YAML.
- The equations can all be easily parsed by SymPy into expressions, and used programmatically.
- The equations refer to some variables, such as R and $CRRA$, which are not defined in the block.

Values for free parameters may be provided in calibration section of the document.

```
calibration: &calibration
  beta: 0.96
  CRRA: 2.0
  R: 1.03
  Rfree: 1.03
  EqP: 0.02
  LivPrb: 0.98
  RiskyStd: 0.1
```

Or these free parameters might be set in other blocks.

2.2 Example: Portfolio choice

Consider another block. This block represents the choice of a consumer to allocate savings to either a risky or risk free asset.

```
block2: &portfolio
  name: portfolio choice
  shocks:
    risky_return: !Lognormal
      mean: Rfree + EqP
      std: 0.2
  dynamics:
    alpha: !Control
      info: a
      lower_bound: 0
      upper_bound: 1
    R: Rfree + (risky_return - Rfree) * alpha
```

This corresponds to the following equations:

$$\begin{aligned}
 \sigma_\eta &= \log((0.2)^2 / (R_{free} + E)^2 + 1) \\
 \mu &= \log(R_{free} + E) - \sigma_\eta / 2 \\
 \eta &\sim \text{Lognormal}(\mu, \sigma_\eta) \\
 \alpha &\in [0, 1] \\
 R &= R_{free} + (\eta - R_{free})\alpha
 \end{aligned} \tag{2}$$

In this block, R , the rate of return, is set endogenously.

3 SEQUENCING

Blocks can be composed into sequences in order to create multi-stage dynamic programming problems.

```
model:
```

```

blocks:
  - *consumption
  - *portfolio
  - twist:
    a: k
calibration: *calibration
structure: !Dynamic
discount: beta

```

This markup introduces a few additional elements:

- We have specified that this is *dynamic* problem, which means that the blocks are repeated with an *infinite horizon*. The agent solving the model *discounts* future utility at a rate β
- Because it's dynamic, the end-state of a needs to be renamed to the beginning-state k . We call such a (very simple) block a *twist*.

The two blocks above compose into a single problem, which can be characterized by the following Bellman equation:

$$\begin{aligned}
 v(k, R) &= \beta \mathbb{E}_\theta \left(\max_c u(c) + \mathbb{E}_\eta [\max_\alpha v(a, R')] \right) \\
 \theta &\sim \text{Lognormal}(1, 0.1) \\
 b &= Rk \\
 m &= b + \theta \\
 a &= m - c \\
 c &< m \\
 u(c) &= \frac{c^{1-\rho}}{1-\rho} \\
 \sigma_\eta &= \log((0.2)^2 / (R_{free} + E)^2 + 1) \\
 \mu &= \log(R_{free} + E) - \sigma_\eta / 2 \\
 \eta &\sim \text{Lognormal}(\mu, \sigma_\eta) \\
 \alpha &\in [0, 1] \\
 R' &= R_{free} + (\eta - R_{free})\alpha
 \end{aligned} \tag{3}$$

This problem can then be passed to many different algorithmic solvers.

4 OTHER DIRECTIONS

We anticipate extending this language to represent a very wide range of models, including models with the following elements:

4.1 Aggregate state

What we have shown is a microeconomic problem faced by a single agent. Macroeconomics concerns what happens when many microeconomic agents such as the portfolio consumer agent interact through a market mechanism.

For example, a macroeconomic model might have wages and the rate of return depend on the total amount of labor and savings provided to the economy by the agents.

We anticipate introducing blocks for operations that span over all the agent states.

4.2 Age varying dynamics

Realistic agents experience different conditions (wage changes, health risks, and so on) as they age.

This means that many calibrated parameters need to be indexed by agent state, such as an incrementing age state.

There are also models in which transitioning between blocks is not deterministic.

4.3 Agent mortality

Birth and death are very important parts of a realistic consumer's life and decision-making process.

We will include in the model ways of configuring the mortality, birth, and inheritance mechanisms at work in the model.

4.4 Multiple agent roles

Often economic models have only one kind of agent: households, or firms. However, realistically, there are many different kinds of agents at work in the economy, and a complex model can include households, firms, banks, and many other kinds of actors.

Introducing more agents can bring a game-theoretic complexity to models. The techniques for working with these kinds of models are worked out more in Agent-Based Modeling (ABM) and Multi-Agent Systems (MAS) disciplines than in Economics per se. But as computational methods advance, the convergence of these fields is inevitable.

4.5 Endogenous and irrational discount factors

Discount factors need not be constant and, indeed, realistically human beings are hyperbolic, not exponential, discounters of future reward.

5 WHERE TO FIND THIS LANGUAGE

We are implementing a reference implementation of an interpreter for this language as part of HARK 1.0.

<https://github.com/econ-ark/hark>